Özgür Kurul

Senior Software Engineer | Full Stack & AI-Native Development

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Languages: Turkish (native), English (professional)

PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years building high-volume, distributed, real-time systems in the C#/.NET ecosystem, delivering backend, web and mobile end to end. Built business-critical microservices and event-driven order and integration flows at international platforms such as Yemeksepeti and Delivery Hero Tech Hub. Currently at Drivee, driving the architecture of an IoT platform processing 25 million data packets a day from 10,000 vehicles and devices, and setting technical direction for a team of six.

Recent focus: AI-native development. Embedded Claude Code, GitHub Copilot and MCP-based agent workflows into every stage of the development cycle, removed a roughly one-day pull-request review wait by handing first-pass review to AI, and shipped LLM-powered product features with the Gemini API. Combines deep experience in .NET, PostgreSQL, Redis, Kafka and asynchronous messaging with React, Next.js and React Native: an engineer who doesn't just call AI models, but turns them into resilient, end-to-end products.

EXPERIENCE

Drivee Technology

February 2024 – Present

Senior Full Stack Engineer · Istanbul, Türkiye

- **End-to-end ownership and technical direction:** Delivered the backend, web and mobile applications of a platform processing 25 million data packets a day from 10,000 vehicles and devices, from architecture through production. Set technical direction for a six-person cross-platform team; established Scrum/sprint practices and drove delivery through code review and mentoring.
- **AI-native development:** Embedded Claude Code and agentic CLI workflows into the development cycle. Used project-specific rule files to enforce code standards while shortening onboarding and feature development time; accelerated database schema work, API documentation, end-to-end test generation and GitHub workflows with agents on PostgreSQL and GitHub MCP servers.
- **AI-assisted code quality:** Removed a roughly one-day pull-request review wait by handing first-pass review to GitHub Copilot Review, leaving the team only a final check before release. Sped up refactoring of legacy modules with Copilot and, together with SonarQube, systematically raised unit and integration test coverage.
- **Architecture transformation:** Broke a monolith made up of a single web application and a TCP application into Kafka-backed producer/consumer, Hangfire job, SignalR WebSocket and Web API services, untangling dependencies; upgraded services to .NET 10. Applied DDD, Clean Architecture, Vertical Slicing and an API Gateway.
- **Event-driven data flow:** Built an event-driven backbone with the Outbox Pattern, Debezium (CDC) and Kafka that ensures every event is processed reliably and streams device data to real-time map views without interruption.
- **Web and mobile:** Built the web portal with React and TypeScript, an SEO-focused landing page with Next.js, and the iOS/Android apps with React Native and Expo.
- **CI/CD (backend + mobile):** Set up quality gates in GitHub Actions (code standards, SonarQube, Copilot Review, unit tests) and Docker build, deploy and rollback pipelines in GoCD. On mobile, produced iOS/Android builds with EAS Build, shipped test builds via TestFlight and Firebase App Distribution, and managed App Store and Google Play releases semi-automatically with EAS.
- **IoT device integration:** Integrated vehicle tracking devices (Teltonika Codec 8, JT808, Armoli and other vendors' hex protocols; RFID, CAN bus), panic button, fuel, door and temperature sensors, and third-party fuel providers (Total, Petrol Ofisi); built device communication over TCP/IP and MQTT (Mosquitto) and handled technical communication with international hardware vendors.
- **Video telematics and alerts:** Integrated live video, driver cameras and AI cameras (CMSv6, Jimi TrackSolidPro); delivered seatbelt, smoking and crash-detection alerts to users within seconds.
- **Cold chain and critical alerting:** Built a high-availability system that monitors cold storage rooms and refrigerators across 80 branches of KFC, HD İskender and Pidem and raises rule-based alerts. Enabled real-time incident response through two-way SMS, automated voice calls and call-center routing.
- **AI and GIS product features:** Built “Drivee AI”, which interprets per-company fleet summary data with the Gemini API to give corporate fleets analysis and insights. Built routing, geofencing and geocoding services with Google services, OpenStreetMap (Overpass, Nominatim), PostGIS and NetTopologySuite.
- **Infrastructure and observability:** Set up geo-redundancy with MS SQL Server Always On and master/replica topologies for Redis and MongoDB. Protected cloud and on-prem Docker environments with WAF/CDN; monitored with Elastic APM and Elasticsearch.

Halkbank

October 2023 – February 2024

Senior Backend Engineer · Istanbul, Türkiye

- Built secure backend microservices for the bank's “Child and Campus Banking” mobile app and migrated its banking features onto the existing API Gateway.
- Developed high-performance, resilient APIs for financial transactions, meeting banking security and compliance standards.

Delivery Hero (Yemeksepeti · Delivery Hero Tech Hub)

August 2021 – October 2023

Senior Software Engineer · Istanbul, Türkiye

Delivery Hero Tech Hub (Integration Hub) April 2022 – October 2023

- In the 25-person Integration Hub team, built a sandbox test environment for companies integrating with Yemeksepeti and Delivery Hero, designed for roughly 750 integrators in Türkiye and 1,500 in Asia-Pacific to run their tests after API migrations.

- Developed integration services and Lambda functions with Go, .NET and AWS (Lambda, SQS, DynamoDB); ran services on Kubernetes and monitored them with Datadog.
- Co-authored the project RFC with the team and increased unit test coverage.
- Analyzed integration needs with Delivery Hero product teams across countries and provided technical support to integrator companies, working in a fully English-speaking, distributed environment.

Yemeksepeti (Yemeksepeti Mahalle) August 2021 – April 2022

- Built the order and campaign microservices of the "Yemeksepeti Mahalle" quick-commerce platform from scratch with .NET and PostgreSQL, handling tens of thousands of orders a day, and played an active role in its launch.
- Implemented the full order lifecycle, from placement to delivery, and its notification flows as event-driven processes on AWS SNS/SQS; ran services on Kubernetes.
- Identified latency hotspots with Datadog traces and added Redis caching layers to improve response times; verified through Locust load tests that the system handled 500,000 requests without issues.

DeFacto

August 2020 – August 2021

Software Engineer · Istanbul, Türkiye

- Developed core features of the global omnichannel e-commerce project, spanning online order management, automated courier assignment and return logistics.
- Improved marketplace, shipping and courier integrations, increasing overall system efficiency.

Novartz

January 2016 – July 2020

Software Engineer · Adana, Türkiye

- **IoT vehicle tracking:** Owned the "Drivee" live vehicle tracking project and established its initial architecture: live tracking, geofencing and driver-scoring algorithms built on TCP/IP socket programming and real-time data streaming.
- **Data analytics:** Built an automated data extraction and reporting system for Starbucks vending machines (Shaya Group).
- **Enterprise solutions:** Delivered 15+ CRM, CMS and corporate portal systems for B2B clients, integrating bank payment gateways and ERP systems such as LOGO and E-Flow.
- **E-commerce:** Delivered the MetalAVM e-commerce platform end to end.

SKILLS

AI-native development: Claude Code, GitHub Copilot and Copilot Review, agentic CLI workflows, MCP (PostgreSQL, GitHub), project-specific rule files, Gemini API integration, n8n

Backend: C#, .NET 10, ASP.NET Core Web API, REST, EF Core, Dapper, SignalR, Hangfire, Quartz, Go

Frontend and mobile: React, Next.js, TypeScript, JavaScript, React Native, Expo

Architecture: Microservices, Domain-Driven Design (DDD), Event-Driven/Async Messaging, Clean Architecture, Vertical Slicing, API Gateway, Outbox Pattern, caching strategies for distributed systems

Databases and GIS: PostgreSQL, PostGIS, MS SQL Server, Redis, MongoDB, NetTopologySuite, Google Maps API, OpenStreetMap (Nominatim, Overpass)

Messaging and IoT: Kafka, Debezium, RabbitMQ, AWS SQS/SNS, Lambda, MQTT (Mosquitto), TCP/IP, WebSocket, device protocols (Teltonika Codec 8, JT808, CMSv6, CAN bus, RFID), video telematics, IVR calling, SMS delivery and tracking

DevOps and CI/CD (backend + mobile): Docker, Kubernetes, GitHub Actions, GoCD, Expo EAS, TestFlight, Firebase App Distribution, Nginx, Kong, HAProxy, WAF, CDN, SonarQube

Observability: Elastic APM, Elasticsearch, Datadog, Graylog, Health Checks, Locust (load testing)

Ways of working: Agile/Scrum, Jira, technical direction and mentoring, system design

EDUCATION

Çukurova University, Adana, Türkiye — B.Sc., Computer Engineering 2012 – 2016

AGH University of Science and Technology, Kraków, Poland — Computer Science, Erasmus+ exchange programme 2015

Osmaniye Korkut Ata University, Osmaniye, Türkiye — Associate Degree, Computer Programming 2010 – 2012

COURSES & CERTIFICATIONS

- GDPR Compliance (2024)
- Cybersecurity Awareness (8 hours, 2023): phishing attacks, malware, IoT security
- Secure Code Warrior (8 hours, 2022 and 2023): secure coding
- Innovation Management Training (24 hours, 2020): modelling and workshops for corporate innovation strategies
- BGA Cyber Security Training (48 hours, 2019): penetration testing, DDoS mitigation, web application security, vulnerability analysis